

Business Data Analysis for Ecovolt Energy

by

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Table of Contents

Table of Contents	1
Introduction	2
1. Context and Background	3
1.1 Description of the Organisation.....	3
1.2 Ecovolt's Sustainability Targets	3
1.3 Collaboration with BIAS (Bright Insight Analytics Services)	3
1.4 Indicators to measure Ecovolt's progress	4
2. Dataset Identification and Description	5
2.1 Dataset Selection and Qualitative Analysis	5
2.2 Preliminary Analysis of Issues in Dataset	6
3. Data Preparation and Outlier Treatment.....	7
3.1 Data cleaning and processing	7
3.2 Outliers, Missing Values & Quantitative Description of Dataset Characteristics	9
4. Data analysis and Visualization.....	12
4.1 Building Interactive Dashboards using Tableau	12
4.2 Graphs Used and Justification of Choice	14
4.3 Analysis and Findings.....	15
4.4 Recommended Action to Client.....	19
5. Conclusion	19
6. References & Use of AI Systems.....	20
Datasets Created	23
Kaggle Notebooks	23

Introduction

Bright Insight Analytics Services (BIAS) specializes in providing organizations with data-driven insights to monitor and track their progress toward achieving the United Nations Sustainable Development Goals (SDGs) ([United Nations, n.d.](#)). The company serves a wide range of clients, including public, nonprofit, and private sector organizations. BIAS's focus is on creating interactive dashboards that allow organizations to assess their sustainability performance using reliable, open data sources.

The current project is in collaboration with Ecovolt Energy, a leading energy provider in Ireland, whose mission is to contribute significantly to SDG 7: Affordable and Clean Energy ([United Nations, n.d.](#)). With Ireland's commitment to ensuring access to clean and affordable energy for all by 2030, Ecovolt seeks to assess its performance against this goal. By leveraging data from sources like Eurostat, the official statistical office of the European Union ([Eurostat, n.d.](#)), BIAS has developed an interactive dashboard to track key sustainability indicators, such as energy consumption, renewable energy share, and energy affordability across Ireland. The insights generated from this dashboard will support Ecovolt in evaluating its progress and making datainformed decisions to achieve SDG 7 targets.

1. Context and Background

1.1 Description of the Organisation

Ecovolt Energy¹, established in 1998 and headquartered in Cork, is a leading electric energy production company managing Ireland's electricity grid. The organization's mission is to provide reliable, sustainable energy while reducing carbon footprint to combat climate change. Serving over 120,000 homes and businesses supplying more than 70% of Ireland's total energy needs, comprising heating, electricity generation, and transport needs. Ecovolt is committed to a sustainable energy approach of decarbonizing electricity processes and ensuring that over 80% of homes have access to affordable, clean energy by 2030, with a 60% reduction in carbon emissions and a fourfold increase in renewable electricity generation by focusing on solar and wind energy projects. By aligning its operations with these goals, Ecovolt aims to reduce the energy dependency on external imports and contribute significantly to the United Nations SDG7 ([United Nations, n.d.](#)).

1.2 Ecovolt's Sustainability Targets

The United Nations Sustainable Development Goal 7 (SDG7) aims to ensure access to affordable, reliable, and sustainable modern energy solutions for all ([United Nations, n.d.](#)). Ecovolt plays a pivotal role in providing clean and sustainable energy to many parts of Ireland, with a keen focus on two key targets under SDG7: ensuring universal access to affordable electricity (SDG 7.1) and increasing the share of renewable energy (SDG 7.2) in the energy mix ([The Global Goals, 2024](#)). Ecovolt aims to achieve these sustainability objectives by 2030, with a major focus on decarbonizing its electricity grids and increasing renewable energy generation.

1.3 Collaboration with BIAS (Bright Insight Analytics Services)

In collaboration with BIAS, Ecovolt seeks to employ advanced analytics and visualization techniques to achieve its sustainability goals. BIAS has developed a visual dashboard that provides stakeholders and consumers with a clear understanding of Ireland's progress towards its sustainability targets compared to other EU countries. These visualizations include features such

¹ Ecovolt is a fictional organization created just for the sole purpose of this assignment and Assuming 70% of Ireland's Electricity is supplied by Ecovolt energy.

as tracking renewable energy usage, identifying energy import trends, and analyzing energy infrastructure efficiency.

Ecovolt intends to use this dashboard to monitor its progress as one of Ireland's major energy agencies. Insights from the dashboard will enable stakeholders and management to make informed decisions, further enhancing operational efficiency and community impact.

1.4 Indicators to measure Ecovolt's progress

Being a major player in Ireland's energy sector, Ecovolt Energy significantly influences the country's sustainability goals. We selected the following indicators ([Central Statistics Office, 2022](#)) to measure and analyze Ecovolt's sustainable approach.

Target	Indicators	Description	Units
Energy Consumption (SDG 7.1)	Primary Energy Consumption (SDG_07_10)	Tracks Total energy needs including non-energy needs like Transmission Losses and Energy Consumed by the power grid itself	MTOE (Million tonnes of oil euuivalent)
	Final Energy Consumption (SDG_07_11)	Focuses on energy used by endusers, excluding sector usage and losses.	MTOE (Million tonnes of oil euuivalent)
Energy Supply (SDG 7.2)	Share of Renewable Energy (SDG_07_40)	Reflects the percentage of renewable energy in total consumption, considering grid losses and self-consumption by power plants.	% (Percentage)
	Energy Import Dependency by Products (SDG_07_50)	Assesses the percentage of energy needs met by imports, crucial for understanding energy security.	% (Percentage)

Access to Affordable Energy (SDG 7.1)	Population Unable to Keep Home Warm (SDG_07_60)	Quantifies the percentage of the population unable to maintain warm housing due to poverty, based on income surveys.	% (Percentage)
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Table 1: List of SDG targets and indicators selected

The above-mentioned indicators (see Table 1) serve as metrics for Ecovolt's sustainability targets. SDG_07_10, and SDG_07_11 quantify the energy demands across various sectors in Ireland, Ecovolt Energy is supplying 70%** of this energy through various sources including oil, natural gas, wind, hydro, and biomass. Indicators SDG_07_40 and SDG_07_50 focus on the share of renewable energy produced and the dependency on imported raw materials, highlighting the need for sustainable domestic energy production. Finally, SDG_07_60 assesses energy poverty and public well-being, aiding Ecovolt in making informed policy decisions and acting as a social safety net.

2. Dataset Identification and Description

2.1 Dataset Selection and Qualitative Analysis

The datasets (see Table 2) chosen for this project come from Eurostat, the official statistical office of the European Union. It is a reliable public database known for its rigorous standards and will be used to analyze cross-country trends related to SDG 7 ([Eurostat, n.d.](#)). The variables in these datasets focus on critical aspects of energy production, consumption, renewable energy usage, and affordability, offering valuable insights into the progress of SDG 7 across the 27 EU member states, which makes them highly relevant for cross-country comparisons, including Ireland. Analyzing these datasets is significant to Ecovolt because 70% of Ireland's total energy is sourced from the company. The invaluable insights from studying, and tracking Ireland's SDG7 progress to the EU will directly contribute to Ecovolt's sustainability targets.

The datasets' information is shown in Table 2 below and they can be classified into three categories, each relevant to energy production and consumption, renewable energy, and

affordability.

Name	Variables	Version/Last updated on	Data collection method	Data type
sdg_07_10: Energy Production	sdg_07_10	05/09/24	Energy Balances (national energy reporting)	Ratio
sdg_07_11: Energy Consumption	sdg_07_11	05/09/24	Energy Balances (national energy agencies, utility data)	Ratio
sdg_07_40: Renewable Energy Share	sdg_07_40	19/09/24	Energy Balances (national renewable energy data)	Ratio
sdg_07_50: Energy Import Dependency	sdg_07_50	15/05/24	Energy Balances (national trade and customs data)	Ratio
sdg_07_60: Energy Poverty	sdg_07_60	04/10/24	Surveys - EU-SILC Data Collection	Ratio

Table 2: Overview of Eurostat Datasets Related to Energy Indicators, SDG 7
[\(Eurostat, 2024, Statistics Explained\)](#)

The datasets sdg_07_10 and sdg_07_11 provide insights into variations in energy production and energy consumption by end users, respectively, from 2000 to 2022. They offer valuable information on energy efficiency, transmission losses, and the energy consumed by power plants within a country [\(Eurostat, 2024, sdg_07_10, sdg_07_11\)](#).

Next, the datasets sdg_07_40 and sdg_07_50 illustrate the share of renewable energy in gross final energy consumption and the extent to which countries rely on imports to meet their total energy needs. These datasets are available from 2004 to 2022 and 2000 to 2022, respectively. Insights from these datasets will help assess the status of a nation’s renewable energy sector and its dependency on external energy sources [\(Eurostat, 2024, sdg_07_40, sdg_07_50\)](#).

Finally, the dataset sdg_07_60 measures the share of the population unable to afford to keep their homes adequately warm due to poverty from 2003 to 2023, offering valuable insights for tracking the progress of energy affordability issues (Eurostat, 2024, sdg_07_60).

2.2 Preliminary Analysis of Issues in Dataset

Several issues were identified in the datasets that could affect the analysis if not addressed properly.

First, the major issue is that the sdg_07_60 dataset has missing data for specific years and countries. These missing data points can disrupt the analysis, especially for countries with significant energy consumption, potentially skewing regional trends and overall conclusions. To address this, targeted imputation strategies—such as interpolation based on neighboring data— will be necessary, or in some cases, the removal of incomplete entries may be more appropriate [\(Bhandari, P., 2023\)](#).

Second, potential outliers were detected across all datasets. These outliers could distort averages, correlations, and trends, making the analysis unreliable. Unusually high or low values could affect trend analysis, so outlier detection using methods like the Z-score or IQR and determining whether to remove or adjust these values is essential ([Levine et al., 2019](#)). While outliers may reveal interesting anomalies, they must be justified or handled carefully to avoid skewing results ([Tan et al., 2019, p. 40](#)).

Third, temporal inconsistencies were identified, particularly with datasets *sdg_07_40*, which starts in 2004, and *sdg_07_60*, which begins in 2003 and includes an additional year, 2023. In contrast, the other datasets span from 2000 to 2022. These discrepancies may complicate comparative timeseries analysis. To resolve this, we should either truncate the datasets to align their timeframes or use extrapolation to fill in the missing years based on available data. This approach also helps address the previously mentioned missing data issues ([Bevilacqua-Linn et al., 2024](#)).

These measures will ensure the datasets are consistent and reliable for further analysis.

3. Data Preparation and Outlier Treatment

3.1 Data cleaning and processing

The five datasets we downloaded from Eurostat, each representing a different indicator, didn't match the website format and included extra text, empty rows, and columns. We merged them into an Excel file ([Nulu, 2024a](#)), and using a Python script² ([Nulu, 2024c](#)), we removed unnecessary text, and non-EU countries, as our focus is the EU. After confirming all EU countries were represented, we saved each dataset into separate sheets within a single Excel file ([Nulu, 2024b](#)).

Next, we created a new script ([Nulu, 2024d](#)) and scanned for missing values in the datasets. The results showed that, except for Sheet *SDG_07_60* (*Population unable to keep home adequately warm by poverty status in Percentage*), no other sheets had missing data:

² Please refer to the Kaggle notebooks in references for details

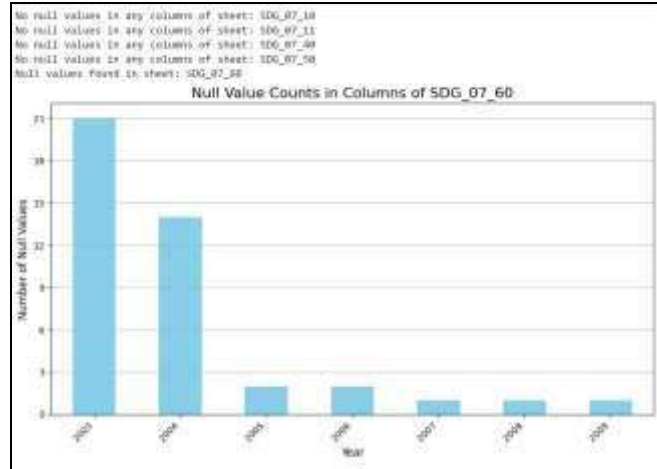


Figure 1: Missing Values Graph shown by year

Figure 1 above revealed many missing values in 2003 and 2004 (21 and 14, respectively). While we could have filled this country-by-country, doing so would rely solely on estimates, which could introduce bias given the significant number of missing values, so we decided to drop those two years ([Bevilacqua-Linn et al., 2024](#); [Bhandari, 2022](#); [Subhak, 2023](#)).

After that, we were left with two countries with missing data: Croatia (from 2005 to 2009) and Romania (from 2005 to 2007). Given this, we considered backfilling ([Mudadla, 2024](#)) and mean imputation ([Little & Rubin, 2019](#)) to fill in the missing values, but both methods would lead to repeating the same value. To better represent the data, the missing values were predicted by fitting a second-degree polynomial model, implemented with linear regression ([James et al., 2013](#)). You can see the results in Figure 2 below.

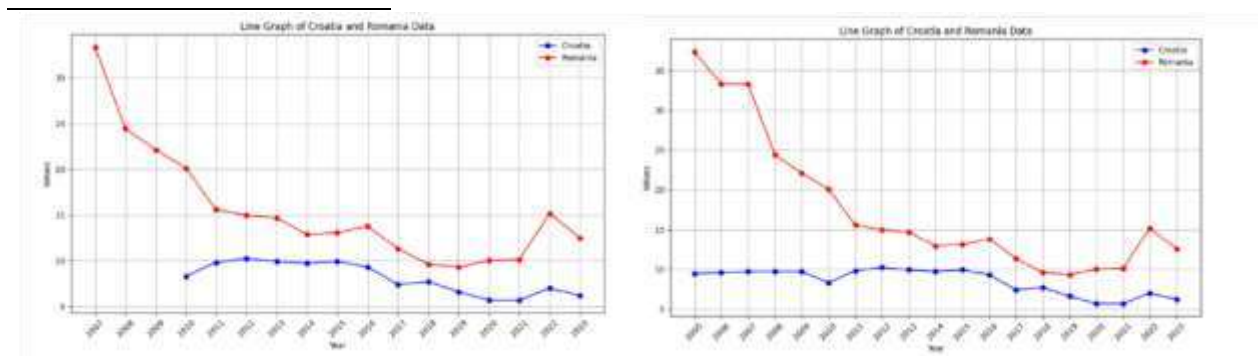


Figure 2: Before (left) and after (right) filling in the missing values

To ensure consistency across all datasets, we removed the years 2000-2004 and 2023, as they were absent from at least one dataset. After making these adjustments, we saved the dataset³ [\(Nulu, 2024e\)](#) and moved on to the next steps.

3.2 Outliers, Missing Values & Quantitative Description of Dataset Characteristics

We ran outlier detection using the IQR method instead of the Z-score. Given the varied socioeconomic indicators across EU countries, the Z-score is unsuitable, as it assumes normally distributed data and could mislabel country-specific variations as outliers. The IQR method, by contrast, better-captured anomalies without overemphasizing deviations from the mean, making it more reliable for our dataset [\(Levine et al., 2020\)](#).

We indexed each sheet by "Country" and flagged any values falling outside the calculated bounds, including only countries with detected outliers in the final results for further review. We then visualized them as shown in Figure 3 on the next page.

³ Please refer to the Kaggle datasets linked in references for details

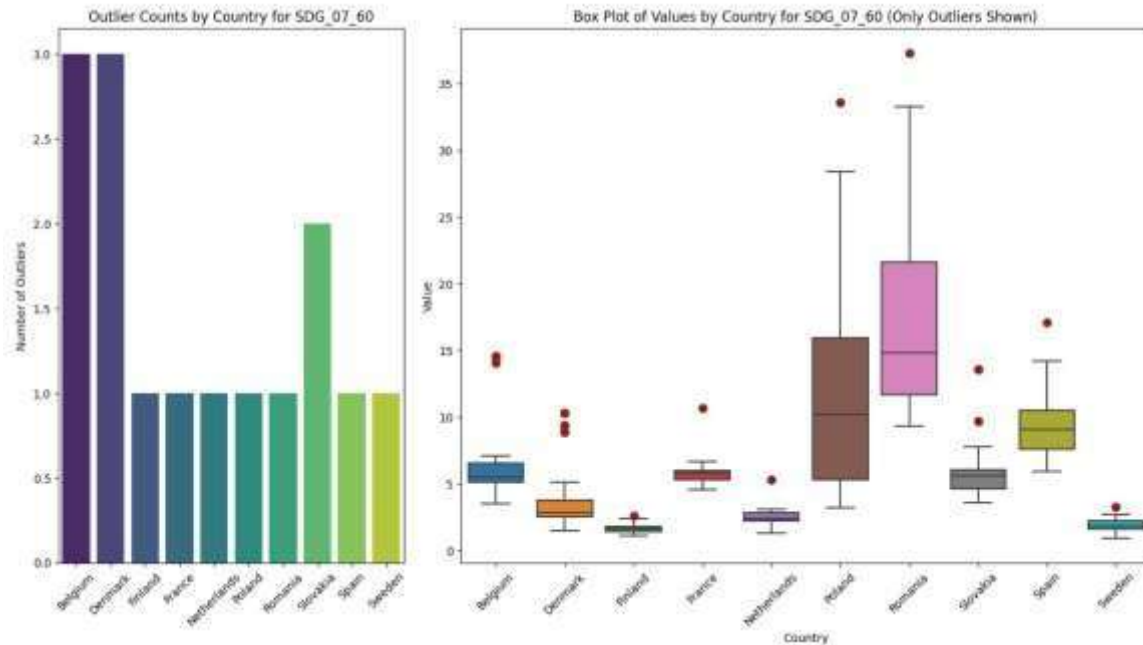


Figure 3: Bar graph displaying the count of outliers per country (left) alongside box plots illustrating outlier values for each country (right) in SDG_07_60 Dataset ([Nulu, 2024d](#))

The outliers detected using the IQR test provide valuable insights into the impacts of economic shifts, pandemics, natural disasters, and other events on a country's energy sector over the past two decades ([Statistics by Jim, 2024](#); [Stats NZ, 2024](#)). We are conducting a case-by-case analysis to determine whether to retain or adjust these outliers, ensuring we accurately reflect true shifts in the data rather than artificially smoothing them out ([Tan et al., 2019, p. 40](#)).

After that, each dataset was summarized with a full suite of descriptive statistics (see Figure 4): mean, median, mode, range, standard deviation, variance, skewness, and kurtosis, to capture central trends, spread, and shape across EU countries from 2005 to 2022 ([Evans, 2020](#); [Levine et al., 2020](#)). These statistics allowed us to analyze the data's overall structure and detect any asymmetries or outliers.

Country	Mean	Median	Mode	Range	Variance	Standard Deviation	Skewness	Kurtosis
Austria	2.7	2.7 [2.7]		2.4	0.5	0.7	-0.1 (Negative)	-0.79 (Platykurtic)
Belgium	6.8	5.5 [5.1, 5.2, 5.6]		11.1	13	3.6	1.54 (Positive)	0.79 (Leptokurtic)
Bulgaria	46.3	42.7 [39.2, 69.5]		47	278.5	16.7	0.21 (Positive)	-1.38 (Platykurtic)
Croatia	8.6	9.5 [9.7]		4.5	2.4	1.5	-0.83 (Negative)	-0.82 (Platykurtic)
Cyprus	26.3	27 [19.2, 19.4]		15.4	26.1	5.1	0.17 (Positive)	-1.25 (Platykurtic)
Czechia	5	5.2 [2.2, 5.2, 6.1]		7.1	4.6	2.2	0.35 (Positive)	-0.71 (Platykurtic)
Denmark	3.9	2.8 [2.7, 2.8, 3.1]		8.8	7.3	2.7	1.5 (Positive)	0.74 (Leptokurtic)
Estonia	2.6	2.7 [1.7, 2.0, 2.1]		3.1	0.6	0.8	0.1 (Positive)	-0.12 (Platykurtic)
Finland	1.7	1.7 [1.7, 1.8]		1.5	0.2	0.4	0.78 (Positive)	0.22 (Leptokurtic)
France	5.9	5.8 [6.0]		6.1	1.7	1.3	2.71 (Positive)	7.7 (Leptokurtic)
Germany	4.7	5 [3.3, 5.5]		4.5	1.6	1.3	-0.15 (Negative)	-0.69 (Platykurtic)
Greece	20.7	18.2 [15.4, 15.7]		20.9	41	6.4	0.53 (Positive)	-1.11 (Platykurtic)
Hungary	9.9	9.6 [5.4]		13.5	15.8	4	0.28 (Positive)	-0.89 (Platykurtic)
Ireland	5.7	4.8 [6.8]		6.6	4.8	2.2	0.64 (Positive)	-1.01 (Platykurtic)
Italy	13.4	11.5 [8.1, 8.3, 8.4]		13.2	16	4	0.43 (Positive)	-1.05 (Platykurtic)
Latvia	15.4	16.6 [16.8]		24.9	53.3	7.3	0.18 (Positive)	-0.97 (Platykurtic)
Lithuania	27.2	27.2 [17.5, 22.4]		18.7	23.8	4.9	0.12 (Positive)	-0.46 (Platykurtic)
Luxembourg	1.4	0.9 [0.9]		3.3	0.8	0.9	0.83 (Positive)	-0.12 (Platykurtic)
Malta	12.1	10.5 [7.6, 7.8]		17.6	33.2	5.8	0.93 (Positive)	-0.47 (Platykurtic)
Netherlands	2.5	2.4 [2.2, 2.4]		4	0.7	0.9	1.81 (Positive)	4.41 (Leptokurtic)
Poland	12.5	10.2 [3.2]		30.4	79.1	8.9	0.99 (Positive)	0.06 (Leptokurtic)
Portugal	26.8	26.9 [17.5]		25.5	66.1	8.1	0.53 (Positive)	-0.82 (Platykurtic)
Romania	17.8	14.8 [33.3]		28	77.7	8.8	1.09 (Positive)	-0.11 (Platykurtic)
Slovakia	6.1	5.6 [4.3, 5.8]		10	5.6	2.4	2.01 (Positive)	3.88 (Leptokurtic)
Slovenia	4.1	4.4 [5.6]		4.4	1.8	1.4	-0.2 (Negative)	-1.28 (Platykurtic)
Spain	9.5	9.1 [8.0]		11.2	7.6	2.8	1.32 (Positive)	1.6 (Leptokurtic)
Sweden	1.9	1.8 [1.7]		2.4	0.4	0.6	0.42 (Positive)	-0.15 (Platykurtic)

Figure 4: Statistics of the SDG_07_60 Dataset ([Nuhu, 2024e](#))

In visualizing the data, we chose to display only the mean and median for Ireland, the highest-performing and lowest-performing countries for each dataset. These countries may vary depending on the indicator. For example, in Figure 5 below, we showed the mean and median for Ireland, Luxembourg (highest-performing), and Bulgaria (lowest-performing) for SDG_07_60. Since the data isn't normally distributed, the median effectively highlights central trends without distortion from outliers, while the mean provides an overall average ([Levine et al., 2020](#)). Displaying these measures enables clearer comparisons across countries and emphasizes trends without overwhelming detail.

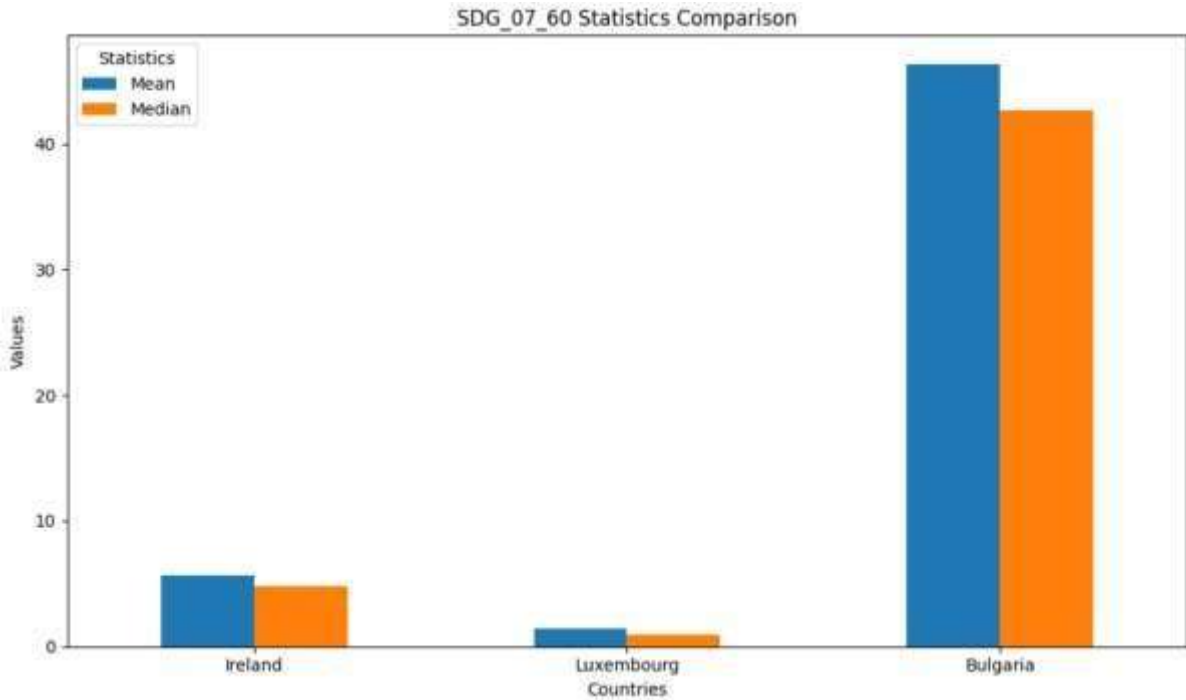


Figure 5: Mean & Median Comparison of Countries of SDG_07_60 (Lower the better) ([Nulu, 2024d](#))

Figure 5 also reflects differences in broader socio-economic and policy variations across the EU. Given these disparities and the influence of population size on energy data, we planned to collect additional population datasets from Eurostat ([Eurostat, 2024](#)). Calculating these indicators per capita would help achieve a clearer, more standardized understanding across countries. But unfortunately, we couldn't find any complete datasets for the population, from Eurostat for all the years we have collected, and we did not want to delete the years any further. As the values on Eurostat were standardized before being converted into percentages, we decided to proceed with the percentages.

4. Data analysis and Visualization

4.1 Building Interactive Dashboards using Tableau

Once all our datasets were prepared, we imported them into Tableau for visualization and dashboard creation. To facilitate graphing and ensure flexibility, we used Python to "melt" the dataset, converting the wide format into a long format. This transformation made the data more adaptable for visual analysis. Afterward, the dataset was imported into Tableau, where we verified the data type attributes to ensure consistency and accuracy ([Pandas Documentation, 2024](#)). Following this, we proceeded with the creation of the necessary calculated fields, which are outlined below.

Transmission Losses (MOTE): The energy lost as electricity travels through power lines from power plants to end users due to resistance and inefficiencies in the system.

$$\text{Transmission Losses} = \text{Primary Energy Consumption} - \text{Final Energy Consumption}$$

This equation reflects that Primary Energy Consumption includes total energy (including losses), while Final Energy Consumption is the energy that reaches users.

Energy Affordability (%): This feature represents the percentage of the population able to keep their home warm, indicating those who can afford energy ([Eurostat, 2024, sdg_07_60](#)).

$$\text{Energy Affordability (\%)} = 100 - \text{Percentage Population Unable to keep home warm(\%)}$$

Energy Infrastructure Efficiency (%): This metric represents the percentage efficiency of a country's energy infrastructure over time, reflecting the impact of improvements made to the energy sector.

$$\text{Energy infrastructure efficiency(\%)} = 100 * \frac{\text{Final Energy Consumption}}{\text{Primary Energy Consumption}}$$

Composite SDG Score: This metric represents a country's overall sustainability status, calculated through the weighted aggregation of key SDG-7 indicators ([Newcastle University, 2024](#)).

$$\text{Composite SDG score} = w1 * \frac{\text{Energy Infrastructure efficiency(\%)}}{100} + w2 * \frac{\text{Share of Renewable energy(\%)}}{100} + w3 * \frac{\text{Energy affordability(\%)}}{100} + w4 * \frac{100 - \text{Energy import dependency(\%)}}{100}$$

This indicator works as an overall indicator because normalizing the metrics by dividing by 100 brings them to a common scale (0-1), making them comparable. The weighted average, using weights (w1, w2, w3, w4), allows assigning relative importance to each indicator based on SDG goals ([Codecademy, 2024](#)).

By using equal weights, we ensure all metrics are given the same level of importance in the final score.

$w_1=w_2=w_3=w_4=25$

4.2 Graphs Used and Justification of Choice

Below, in Figure 6, we present the complete dashboard.

To depict the average Composite SDG-7 score for each country, we used a geographical map with a color gradient from green to red. This allowed us to easily identify high-performing (green) and underperforming (red) regions, providing a clear geographic distribution.

For the time series analysis of infrastructure developments in Ireland, the highest- and lowestperforming countries based on Energy Infrastructure Efficiency (%), we chose a line plot. This visualization effectively shows trends over time, allowing us to observe fluctuations in efficiency.

A scatter plot was used to examine the correlation between the indicators Share of Renewable Energy (%) and Energy Import Dependency (%). We further employed a clustered bar chart to visualize the relationship between energy imports and the share of renewable energy, highlighting how increases in renewable energy production (%) correlated with a reduction in energy imports (%).

To compare Energy Affordability (%) across countries, we used a horizontal bar chart. The simplicity and clarity of the bar graph made it an ideal choice for visualizing this single metric, ensuring precision in the representation of data ([WPDDataTables, 2024](#)).

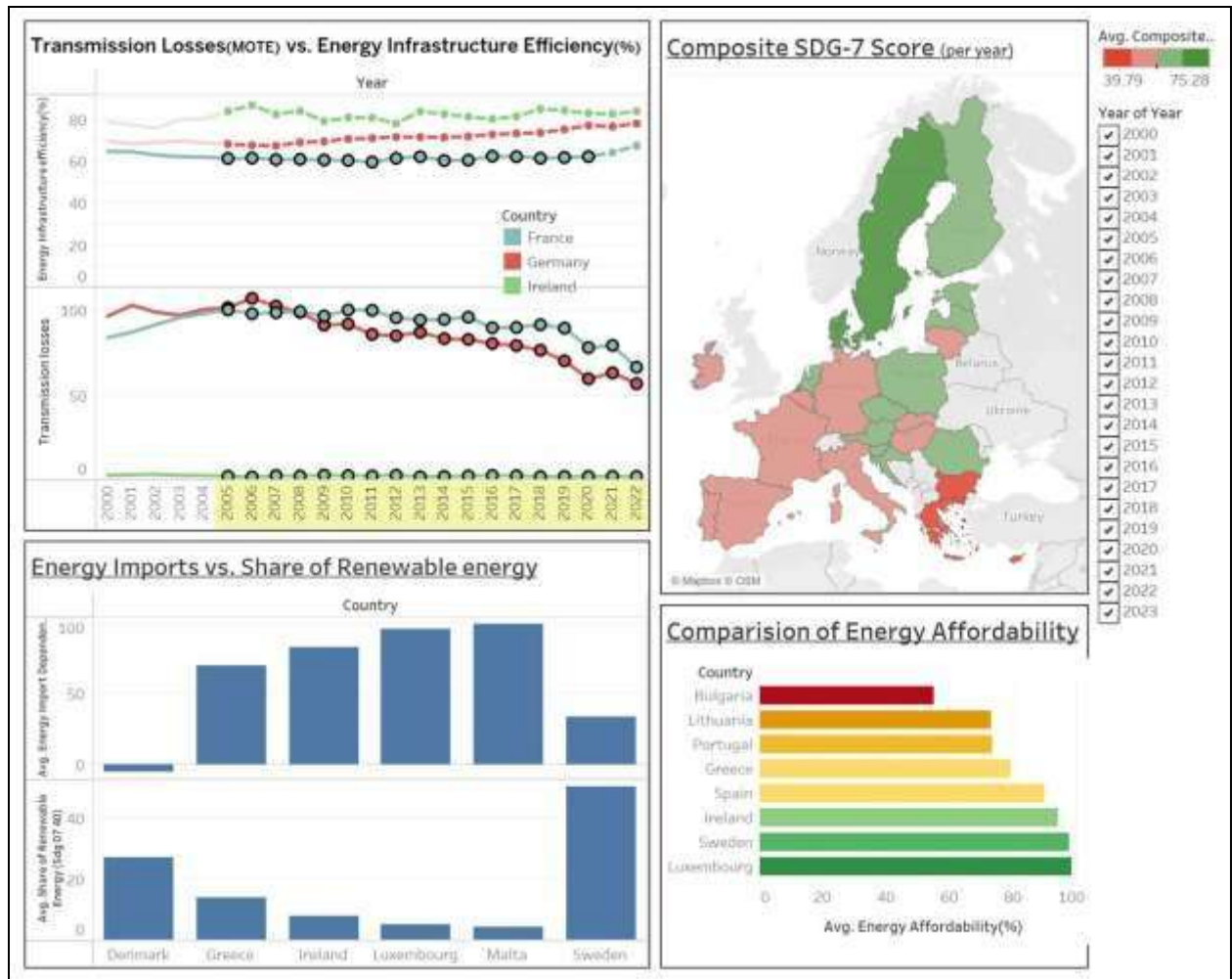


Figure 6: Overall Dashboard Overview

4.3 Analysis and Findings

The correlation factors (R -squared:0.3046 & P -value:0.036) from the polynomial curve revealed a strong relationship between Energy imports (%) and Share of renewable energy (%) (see Figure 7). Indicating that as countries tend to increase the share of renewable sources of energy the dependency on external energy imports decreases simultaneously. As a result, the countries that have concentrated renewable energy sources are less dependent on external energy imports (Frost, 2024).

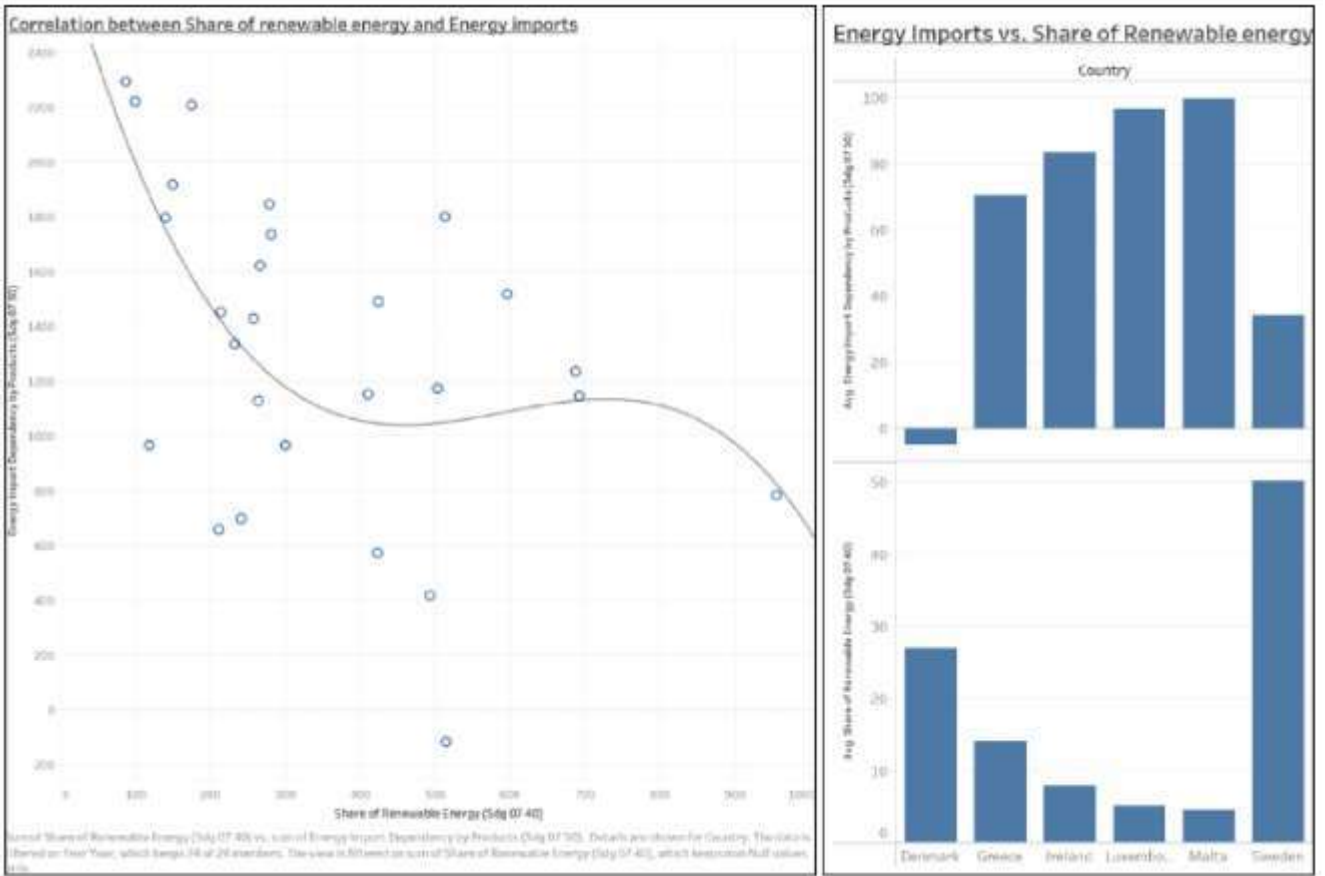


Figure 7: Relationship between Energy Imports and Share of Renewable energy

The stacked line graph (see Figure 8) shows how transmission losses influence the Energy infrastructure efficiency (%), interestingly Ireland managed to maintain almost constant energy efficiency over the past two decades with minor fluctuations in 2006 & 2013. Whereas, Germany's and France's recent reduction in transmission losses likely indicate significant technological upgrades in the energy sector. These two countries account for the major changes in transmission losses compared to the remaining EU countries ([Tableau Software, 2024](#)).

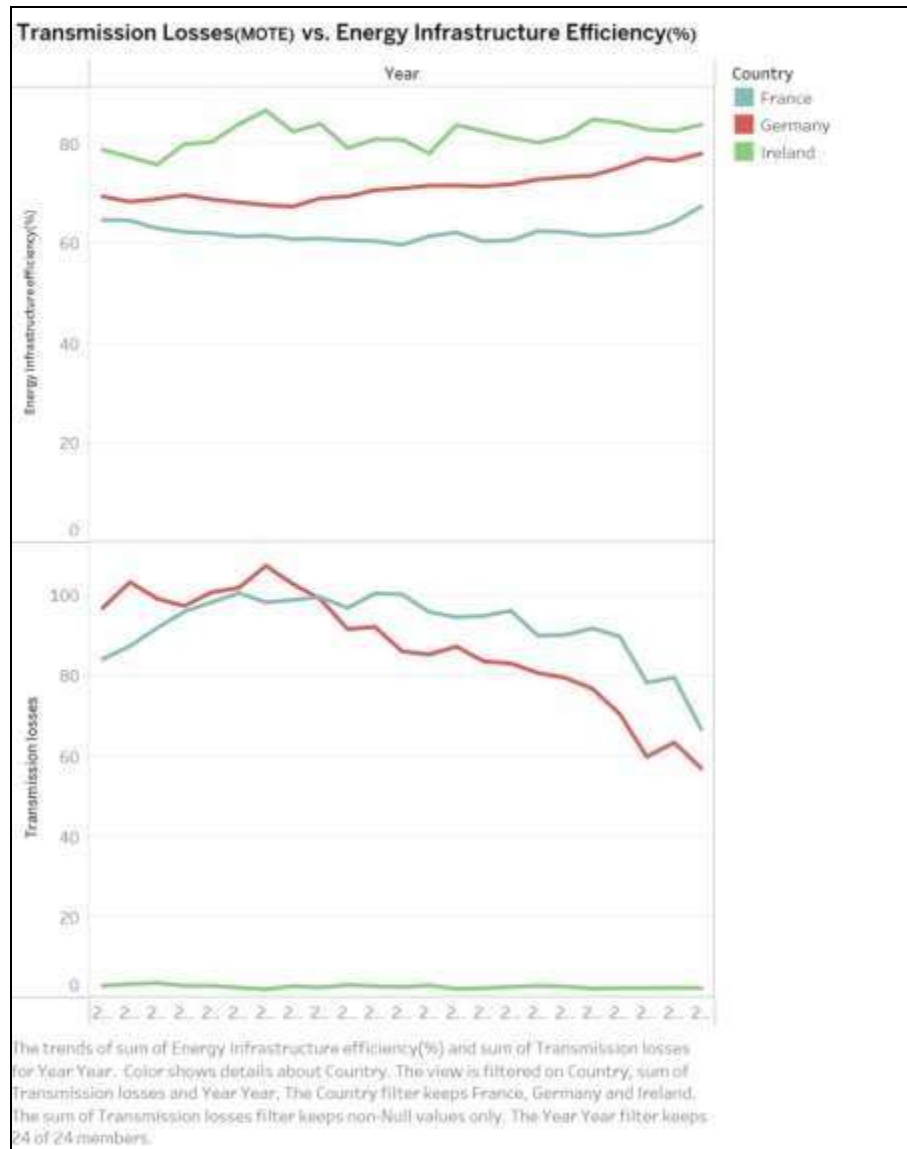


Figure 8: Trends in Energy infrastructure efficiency and Transmission losses

Next, visualizing composite SDG scores revealed interesting facts like, the majority of the EU countries are lagging in SDG-7 goals and some are even drifting away from SDG-7 targets.

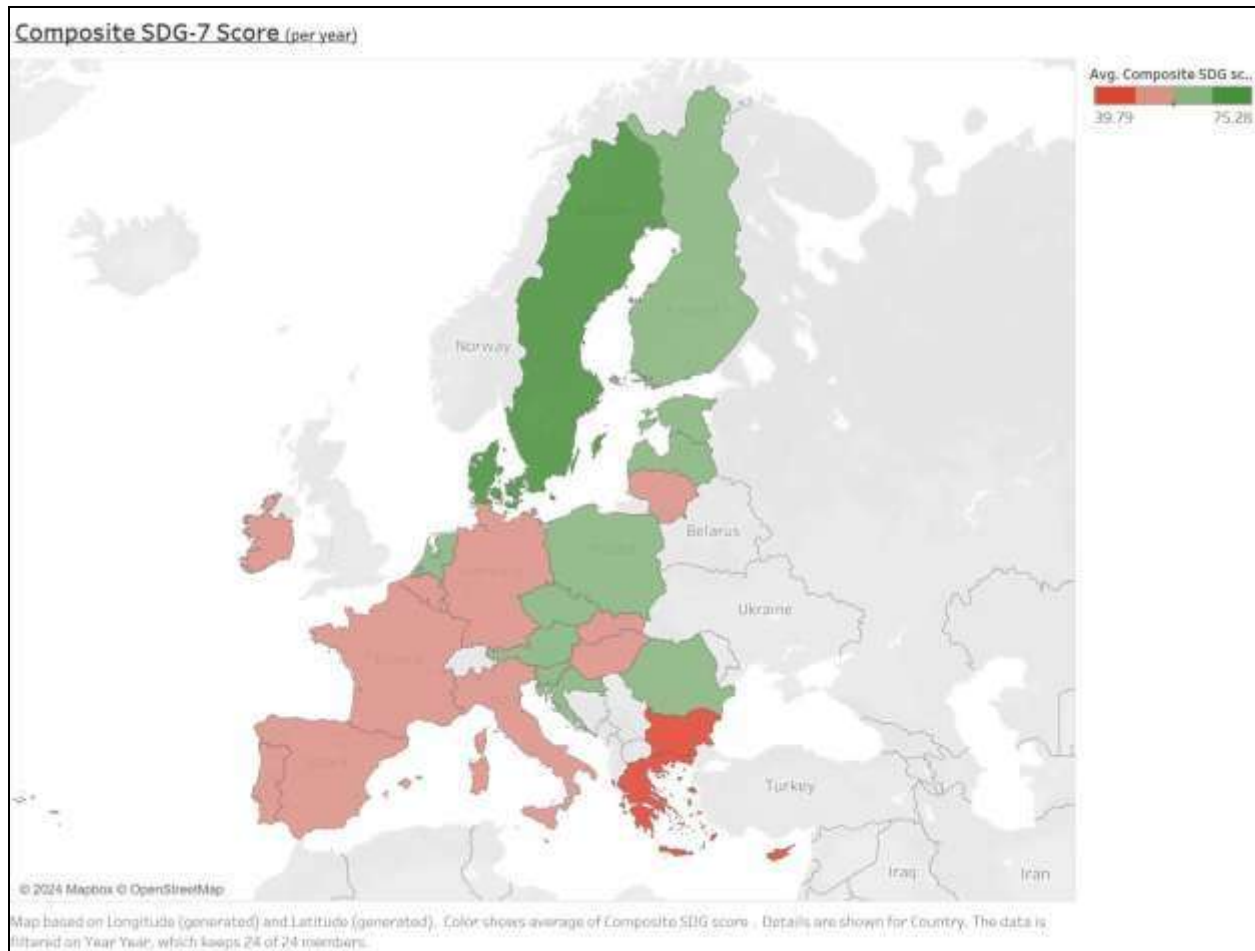


Figure 9: Composite SDG-7 score of all EU countries

Upon further analysis, it was observed that countries in Northern and Western Europe (e.g., Sweden, Finland, Denmark) achieved higher SDG-7 scores, marked in green, compared to those in Eastern and Southern Europe (e.g., Bulgaria, Romania, Greece), which displayed lower scores in red (Figure 9). This disparity reflects the impact of factors such as sustainable policies, efficient infrastructure, economic or political constraints, and more favorable socioeconomic conditions in the former regions.

Bulgaria faces the most significant energy affordability challenges in the EU, with only about 60% of its population able to afford heating their homes (see Figure 10). Lithuania and Portugal also struggle, with approximately only 73-75% of their populations able to cover energy costs.

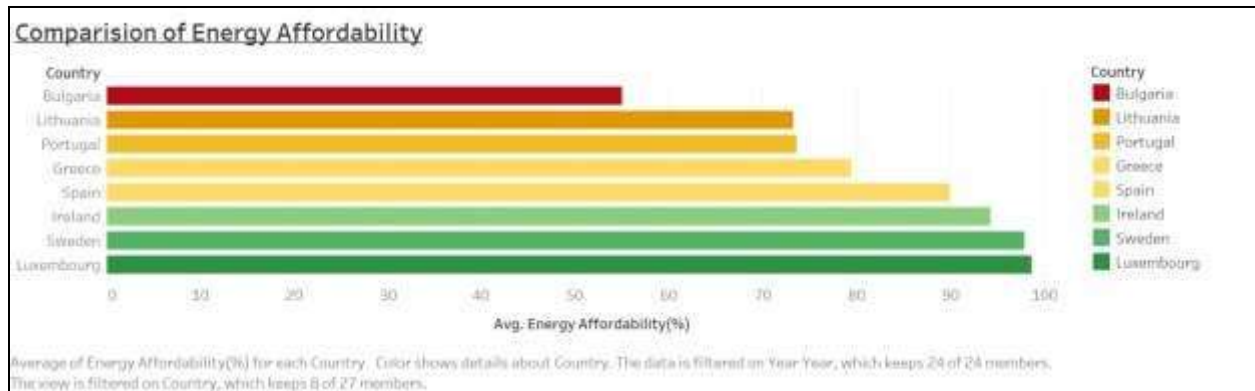


Figure 10: Composite SDG-7 score of all EU countries

In contrast, Ireland, Sweden, and Luxembourg report high energy affordability, with more than 90% of their populations able to afford heating. These disparities across countries are likely driven by factors such as energy prices, income levels, and government policies.

4.4 Recommended Action to Client

Scale Up Renewable Energy Projects: Using Ecovolt’s expertise in the energy sector to expand solar, wind, and other renewable energy facilities across Ireland. Partner with local authorities and communities to develop decentralized energy solutions which can reduce dependency on imports, and align with Ecovolt’s SDG commitment.

Optimize Energy Transmission: Upgrading aging infrastructure with smart grids will minimize transmission losses and boost energy efficiency.

Promote Affordability: New community-friendly initiatives to modify homes with energyefficient solutions, such as insulation and smart energy systems focusing on affordability for lowincome households.

5. Conclusion

In conclusion, the analysis of Ecovolt’s progress toward SDG 7 through the creation of an interactive dashboard offers a clear and actionable understanding of its performance. The use of reliable, open-source data has allowed Ecovolt to monitor key indicators such as energy consumption patterns, the share of renewable energy, and energy affordability. While Ecovolt has made considerable progress in increasing renewable energy generation, challenges remain, particularly in the area of energy affordability for vulnerable populations.

The dashboard provides Ecovolt with valuable insights that will help guide future decisionmaking. By continuing to leverage data visualization tools and refining strategies, Ecovolt can enhance its

efforts toward achieving a sustainable energy future. Through collaboration with BIAS, the company is positioned to drive impactful change, not only in Ireland but also in contributing to global efforts toward achieving SDG 7 by 2030.

6. References & Use of AI Systems

All data and information referenced in this analysis are derived from Eurostat, the official statistical office of the European Union. Eurostat's datasets, methodologies, and explanations are cited appropriately to support all claims and analyses made in this project.

Furthermore, no content generation or analysis was conducted using AI systems; AI tools were only utilized for proofreading and ensuring the clarity of the submission. All external material is fully cited, following the APA referencing style.

1. United Nations. (n.d.). *Sustainable Development Goals*. United Nations. <https://sdgs.un.org/goals>
2. United Nations. (n.d.), Goal 7: Ensure access to affordable, reliable, sustainable and modern energy for all. United Nations Sustainable Development Goals: <https://sdgs.un.org/goals/goal7>
3. The Global Goals. (2024). Goal 7: Affordable and clean energy. The Global Goals. <https://www.globalgoals.org/goals/7-affordable-and-clean-energy/>
4. Central Statistics Office, Goal 7: Affordable and clean energy, Regional EU sustainable development indicators for Ireland, 2022: <https://www.cso.ie/en/releasesandpublications/ep/p-eusdi/regionaleusustainabledevelopmentindicatorsforireland2022/goal7affordableandcleanenergy/>
5. Eurostat. (n.d.). Who we are. Eurostat. Retrieved December 1, 2024, from <https://ec.europa.eu/eurostat/en/web/main/about-us/who-we-are>
6. Eurostat. (2024). Statistics explained - Energy balance - new methodology. Retrieved October 11, 2024, from https://ec.europa.eu/eurostat/statisticsexplained/index.php?title=Energy_balance_-_new_methodology
7. Eurostat. (2024). Primary energy consumption (sdg_07_10). Retrieved September 27, 2024, from https://ec.europa.eu/eurostat/cache/metadata/en/sdg_07_10_esmsip2.htm

8. Eurostat. (2024). Final energy consumption (sdg_07_11). Retrieved September 27, 2024, from https://ec.europa.eu/eurostat/cache/metadata/en/sdg_07_11_esmsip2.htm
9. Eurostat. (2024). Share of renewable energy in gross final energy consumption by sector (sdg_07_40). Retrieved September 27, 2024, from https://ec.europa.eu/eurostat/cache/metadata/en/sdg_07_40_esmsip2.htm
10. Eurostat. (2024). Energy import dependency by products (sdg_07_50). Retrieved September 27, 2024, from https://ec.europa.eu/eurostat/cache/metadata/en/sdg_07_50_esmsip2.htm
11. Eurostat. (2024). Population unable to keep home adequately warm by poverty status (sdg_07_60). Retrieved September 27, 2024, from https://ec.europa.eu/eurostat/cache/metadata/en/sdg_07_60_esmsip2.htm
12. Eppler, M., & Bünzli, F. (2022). How to Talk about Data: Build your data fluency. FT Publishing International.
13. Eurostat. (2024). Statistics explained - EU statistics on income and living conditions (EU-SILC) methodology. Retrieved October 11, 2024, from [https://ec.europa.eu/eurostat/statistics-explained/index.php?title=EU_statistics_on_income_and_living_conditions_\(EU-SILC\)_methodology](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=EU_statistics_on_income_and_living_conditions_(EU-SILC)_methodology)
14. Bhandari, P. (2023, June 21). What Is Data Cleansing? | Definition, Guide & Examples. Scribbr. Retrieved October 8, 2024, from <https://www.scribbr.com/methodology/data-cleansing/>. Topic 15, 16.
15. Levine, D. M. ., Szabat, K. A. ., & Stephan, D. F. . (2020). Business Statistics: A First Course, Global Edition. Chapter 3.
16. Tan, P., Steinbach, M., Karpatne, A., & Kumar, V. (2019). Introduction to Data Mining. Addison-Wesley.
17. Bevilacqua-Linn, M., Machado, R., & N., V. (2024). How can you handle missing values in a time-series dataset? How to Handle Missing Values in Time-Series Data. <https://www.linkedin.com/advice/1/how-can-you-handle-missing-values-time-series-mwe-yc>
18. Subhak, P. (2023, March 2). Handling missing values in dataset: 7 methods that you need to know. Medium. Retrieved from <https://medium.com/@pingsubhak/handling-missing-values-in-dataset-7-methods-that-yo>

[u-need-to-know-5067d4e32b62](#)

19. Mudadla, S. (2024). What is forward and backward filling of imputation in exploratory data analysis (EDA)? Medium. Retrieved from <https://medium.com/@sujathamudadla1213/what-is-forward-and-backward-filling-of-imputation-in-exploratory-data-analysis-eda-74ec33aca4bd>
20. Little, R. J. A., & Rubin, D. B. (2019). Statistical analysis with missing data (3rd ed.). Wiley.
21. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning with applications in R. Springer.
22. Statistics by Jim. (2024). How to remove outliers from your data. Retrieved from <https://statisticsbyjim.com/basics/remove-outliers/>
23. Stats NZ. (2024). Seasonal adjustment and additive outliers during COVID-19. Statistics New Zealand. Retrieved from <https://www.stats.govt.nz/methods/seasonal-adjustment-and-additive-outliers-during-covid-19>
24. Evans, J. (2020). Business analytics (Global ed.). Pearson Education, Limited.
25. Eurostat. (2024). Population change - Demographic balance and crude rates at national level [Data set]. Retrieved from https://ec.europa.eu/eurostat/databrowser/view/DEMO_GIND_custom_2733962/bookmark/table?lang=en&bookmarkId=7084ed24-6b91-4cf3-b90d-d47565593505
26. Pandas Documentation. (2024). pandas.melt function, from <https://pandas.pydata.org/docs/reference/api/pandas.melt.html>
27. Newcastle University. (2024). Weighted averages. <https://www.ncl.ac.uk/webtemplate/ask-assets/external/maths-resources/statistics/descriptive-statistics/weighted-averages.html#:~:text=First%20work%20out%20the%20percentage,the%20weight%20of%20each%20module.&text=Finally%20to%20find%20your%20mark,%2015%200.4%3D0.7.>
28. Codecademy. (2024). Normalization. <https://www.codecademy.com/article/normalization>
29. WPDataTables. (2024). Types of charts. <https://wpdatatables.com/types-of-charts/>

30. Frost, J. (2024). How to interpret R^2 in regression. Statistics By Jim.
<https://statisticsbyjim.com/regression/interpret-rsquared-regression/>
31. Tableau Software. (2024). Scatter plot examples.
https://help.tableau.com/current/pro/desktop/en-us/buildexamples_scatter.htm

Datasets Created

1. Nulu, S. C. (2024a). *Eurostat Dataset SDG 7 - The Original Dataset extracted from Eurostat*. Kaggle. Retrieved from
<https://www.kaggle.com/datasets/srichaitanyanulu/eurostat-dataset-sdg-7>
2. Nulu, S. C. (2024b). *SDG_07_Dataset_Cleaned - The Dataset After Removing Unnecessary Explanatory Text and Formatting Anomalies*. Kaggle. Retrieved from
<https://www.kaggle.com/datasets/srichaitanyanulu/sdg-07-dataset-cleaned>
3. Nulu, S. C. (2024e). Final SDG 07 Dataset and SDG 07 Statistics. Kaggle. Retrieved from
<https://www.kaggle.com/datasets/srichaitanyanulu/final-sdg-07-dataset-and-sdg-07-statistics>

Kaggle Notebooks

1. Nulu, S. C. (2024c). *Eurostat Dataset Cleanup*. Kaggle Notebooks. Retrieved from
<https://www.kaggle.com/code/srichaitanyanulu/eurostat-dataset-cleanup>
2. Nulu, S. C. (2024d). *SDG_07_Dataset Preprocessing*. Kaggle Notebooks. Retrieved from
<https://www.kaggle.com/code/srichaitanyanulu/sdg-07-dataset-preprocessing>